

Language-Defined Attack Semantics for Federated Intrusion Detection

Prateek Gupta
IIT Kharagpur
pg7729600@gmail.com

Prasenjit Betal
IIT Kharagpur
prasenjitbetal.23@kgpian.iitkgp.ac.in

Arindam Dasgupta
IIT Kharagpur
adgkqp@gmail.com

Pabitra Mitra
IIT Kharagpur
pabitra@cse.iitkgp.ac.in

Abstract—Federated intrusion detection systems must operate under severe data heterogeneity, privacy constraints, and evolving attack surfaces. Conventional classifiers force all network flows into a closed set of known attack classes, resulting in overconfident misclassification under uncertainty and distribution shift.

In this work, we develop a framework that formulates detection as a gated decision problem rather than closed-set classification. Each attack class is constructed and grounded in a semantic embedding space derived from large language model descriptions, enabling similarity-based reasoning and calibrated rejection. We introduce class-specific similarity thresholds, ambiguity-aware margin rejection, and federated centroid alignment via exponential moving averages.

Experiments on the NSL-KDD dataset under non-IID client splits demonstrate that COSINE achieves up to 97.8% robust accuracy while maintaining controlled traffic coverage. Ablation studies show that margin-only training maximizes precision, semantic-only training improves coverage, and the combined semantic-margin objective yields the best operational trade-off.

Index Terms—Federated Learning, Intrusion Detection Systems, Open-Set Recognition, Semantic Embeddings, Network Security

I. INTRODUCTION

Intrusion Detection Systems (IDS) are a critical component of modern network security infrastructure [1], providing continuous monitoring and alerting against malicious activity across enterprise, cloud, and edge networks. As networked systems grow in scale and complexity, IDS must operate under increasingly stringent constraints, including limited visibility across administrative domains, strict data privacy regulations, and rapidly evolving threat landscapes. These constraints make it difficult for traditional centralized IDS pipelines to remain effective in real-world deployments [2].

Three key challenges arise: (i) privacy constraints prevent centralized data aggregation [3], (ii) network traffic is highly non-IID across clients [4], and (iii) real-world traffic includes unknown or ambiguous attacks that do not align with predefined classes [5]. These factors limit the robustness of closed-set IDS models.

Federated Learning (FL) has emerged as a promising paradigm for addressing privacy concerns by enabling collaborative model training without sharing raw data. [6] Recent work has explored federated IDS architectures using deep neural networks, autoencoders, and hybrid anomaly detection techniques. [7] Although these approaches successfully

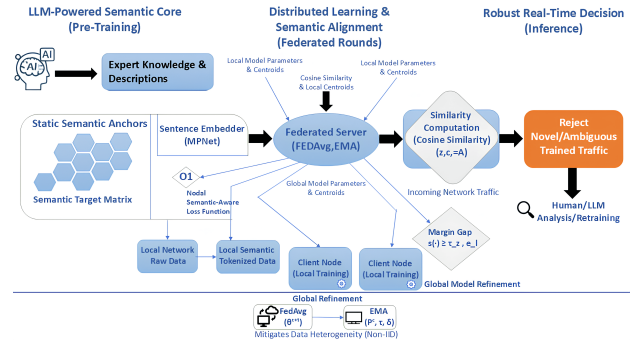


Fig. 1: COSINE system pipeline. Network traffic is locally embedded on clients and trained via federated learning. The global model aligns embeddings to semantic attack prototypes. Final decisions are produced using similarity-based gated rejection with class-specific thresholds.

preserve data locality, they exacerbate uncertainty in heterogeneous client distributions. In particular, most federated IDS systems inherit a closed-set classification assumption, forcing every observed network flow to be assigned to a known attack class [8]. This design choice leads to dangerous overconfidence when encountering novel attacks or boundary cases, where the highest-confidence prediction may still be semantically incorrect.

To address these limitations, we propose a framework named *COSINE* (COllaborative Semantic Intrusion Network Embedding)(Fig. 1), a federated semantic intrusion detection framework that reformulates intrusion detection as a gated decision problem rather than a forced classification task. Instead of relying on discriminative class boundaries, COSINE embeds traffic into a shared semantic space and measures similarity to LLM-derived attack prototypes, enabling reasoning about semantic alignment rather than forcing a label.

COSINE introduces class-specific similarity thresholds and ambiguity-aware margin rejection to model uncertainty at inference time. When the evidence is insufficient or conflicting, the system abstains rather than produces weak predictions. To ensure stability under federated heterogeneity, semantic class centroids are refined using exponential moving averages across training rounds. Experimental results on the NSL-KDD dataset demonstrate that COSINE achieves high robust accuracy while

maintaining controlled traffic coverage, highlighting its effectiveness as a practical, uncertainty-aware federated IDS.

Contributions. We propose a language-defined semantic framework for intrusion detection that models attack classes in a shared embedding space, introduces a similarity-based gated decision mechanism with ambiguity-aware rejection, and develops a federated semantic alignment strategy to handle non-IID data. Experiments demonstrate improved robust accuracy with controlled traffic coverage.

Organization. Section II reviews related work. Section III presents the proposed method. Section IV describes experimental setup. Section V discusses results. Section VI concludes the paper.

II. BACKGROUND AND RELATED WORK

A. Federated Learning for Intrusion Detection

Federated learning enables collaborative intrusion detection without centralizing raw network traffic, preserving privacy across administrative domains. [9] However, non-IID client distributions and partial class visibility significantly degrade closed-set IDS models, motivating uncertainty-aware decision mechanisms.

B. Open-Set Recognition and Rejection in IDS

Prior IDS approaches incorporate rejection through global confidence thresholds or anomaly scores [10]. These mechanisms are brittle under distribution shift and fail to account for class-specific semantic variability, particularly in federated settings with heterogeneous client data.

C. Semantic Representations and Positioning of This Work

Recent advances in natural language processing have enabled the use of semantic representations for cybersecurity tasks, including malware analysis, threat intelligence mining, and vulnerability classification. [11], [12] By grounding technical artifacts in semantic embedding spaces derived from textual descriptions, these approaches allow models to reason beyond purely statistical correlations.

In intrusion detection, semantic information has primarily been used for post-hoc analysis, alert explanation, or rule enrichment. Few systems integrate semantic representations directly into the core detection logic, and even fewer explore their use in federated settings. Existing methods typically decouple semantic reasoning from model training, limiting their ability to influence decision boundaries or uncertainty estimation.

In contrast to prior work that uses language or semantic embeddings as auxiliary attributes for classification or post-hoc explanation, COSINE treats generated descriptions as class-defining semantic primitives [13], [14]. Attack classes are constructed, aligned, and calibrated through language, and all model behavior is constrained by this definition. This shift enables federated alignment under partial class visibility, calibrated rejection grounded in semantic evidence, and explicit modeling of uncertainty—capabilities not addressed by existing federated or LLM-assisted IDS frameworks.

III. MODEL ARCHITECTURE

COSINE is designed as a federated embedding-based intrusion detection framework that explicitly separates *representation learning* from *decision-making*. Given an input network flow, the model maps raw features into a normalized semantic embedding space, evaluates similarity to class-level semantic prototypes, and applies gated rejection based on calibrated confidence criteria. This section describes the full computational pipeline from input to final decision.

A. Semantic Attack Class Construction via Generated Descriptions

Unlike conventional intrusion detection systems that treat attack labels as fixed categorical identifiers [15], COSINE defines each attack class through generated natural language descriptions. For each attack type, a large language model is used to synthesize multiple paraphrased descriptions that capture operational intent and behavior. These descriptions are embedded and aggregated to construct a semantic prototype, which serves as the canonical representation of the class.

This design makes language a first-class modeling primitive: attack classes do not exist independently of their descriptions, and all downstream learning, decision-making, and rejection are governed relative to these language-defined semantic anchors. As a result, COSINE can reason about attack classes that are unobserved at specific clients, enabling federated alignment, zero-shot generalization, and principled rejection under uncertainty [16], [17].

B. Input Representation

Each network flow is represented as a feature vector

$$\mathbf{x} \in \mathbb{R}^{d_x},$$

obtained after standard preprocessing, including categorical feature encoding and normalization. No packet payloads or raw traffic traces are shared across clients, ensuring privacy preservation in the federated setting.

Each client k holds a local dataset

$$\mathcal{D}_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_k},$$

where $y_i \in \mathcal{Y}$ denotes the attack label associated with the flow, and $\mathcal{Y}$ is the set of known attack classes.

C. Embedding Network

The core of COSINE is a neural embedding function [18]

$$f_\theta : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^d,$$

parameterized by θ and shared across all clients via federated learning. The embedding network consists of multiple fully connected layers with nonlinear activations, followed by ℓ_2 normalization to constrain representations to the unit hypersphere:

$$\mathbf{z} = f_\theta(\mathbf{x}), \quad \hat{\mathbf{z}} = \frac{\mathbf{z}}{\|\mathbf{z}\|_2}.$$

This normalization ensures that cosine similarity corresponds to the dot product between embeddings and enables stable geometric reasoning in the semantic space.

D. Semantic Attack Prototypes

Each attack class $c \in \mathcal{Y}$ is represented by a semantic prototype

$$\mathbf{p}_c \in \mathbb{R}^d,$$

derived from textual descriptions generated using large language models (LLMs). For each class, multiple paraphrased descriptions are embedded using a sentence encoder, normalized, and aggregated into a centroid:

$$\mathbf{p}_c = \frac{1}{N_c} \sum_{j=1}^{N_c} \frac{\mathbf{e}_{c,j}}{\|\mathbf{e}_{c,j}\|_2}, \quad \mathbf{p}_c \leftarrow \frac{\mathbf{p}_c}{\|\mathbf{p}_c\|_2}.$$

These prototypes are not auxiliary descriptors but define the semantic identity of each attack class, forming the reference geometry for training, inference, and rejection.

E. Similarity Computation

Given an embedded network flow $\hat{\mathbf{z}}$, COSINE computes cosine similarity scores with respect to all class prototypes:

$$s_c(\mathbf{x}) = \cos(\hat{\mathbf{z}}, \mathbf{p}_c) = \hat{\mathbf{z}}^\top \mathbf{p}_c, \quad \forall c \in \mathcal{Y}.$$

Let $s_{(1)}$ and $s_{(2)}$ denote the highest and second-highest similarity scores, respectively, with the corresponding class labels $c_{(1)}$ and $c_{(2)}$.

F. Training Objective

Clients optimize the embedding network using contrastive objectives that enforce geometric separation between attack classes [19]. For a given sample, a margin-based contrastive loss is defined as:

$$\mathcal{L}_{\text{margin}} = \max(0, m - s^+ + s^-),$$

where s^+ is the similarity to the correct class prototype and s^- is the similarity to a wrong prototype randomly sampled.

To incorporate semantic alignment, COSINE optionally augments this objective with a semantic loss term:

$$\mathcal{L}_{\text{semantic}} = 1 - \cos(\hat{\mathbf{z}}, \mathbf{p}_y),$$

yielding the combined objective:

$$\mathcal{L} = \mathcal{L}_{\text{margin}} + \lambda \mathcal{L}_{\text{semantic}}.$$

G. Federated Optimization

Each client performs local optimization on $\mathcal{D}_k$ and periodically synchronizes model parameters using federated averaging:

$$\theta^{(t+1)} = \sum_{k=1}^K \frac{N_k}{\sum_j N_j} \theta_k^{(t)}.$$

To mitigate semantic drift across clients, global class prototypes are refined using exponential moving averages (EMA) of client-derived centroids:

$$\mathbf{p}_c^{(t+1)} = (1 - \alpha) \mathbf{p}_c^{(t)} + \alpha \bar{\mathbf{p}}_c^{(t)},$$

followed by normalization.

H. Gated Decision Rule

Rather than forcing a classification, COSINE applies a gated decision mechanism [20]. A test sample is accepted as class $c_{(1)}$ only if two conditions are satisfied:

$$s_{(1)} \geq \tau_{c_{(1)}} \quad \text{and} \quad s_{(1)} - s_{(2)} \geq \delta,$$

where τ_c is a class-specific similarity threshold and δ is a margin gap parameter.

The final prediction rule is:

$$\hat{y}(\mathbf{x}) = \begin{cases} c_{(1)}, & \text{if accepted,} \\ \text{REJECT,} & \text{otherwise.} \end{cases}$$

This design allows COSINE to explicitly reject ambiguous or novel traffic rather than producing overconfident misclassifications.

IV. EXPERIMENTAL SETUP

This section describes the dataset, federated configuration, training protocol, and evaluation methodology used to assess COSINE under realistic non-IID and privacy-constrained deployment conditions. The training and optimization workflow of our proposed framework is systematically detailed in Algorithm 1. This procedure highlights the synchronization between local client optimization and the global semantic alignment performed at the server.

A. Dataset and Preprocessing

Experiments are conducted on the NSL-KDD dataset [21], a standard benchmark for network intrusion detection containing both benign traffic and multiple attack categories. Raw network flow records are converted into fixed-length feature vectors using categorical encoding and feature normalization. No packet payloads or raw traffic traces are shared across clients, ensuring compliance with federated privacy constraints.

The dataset is partitioned into three disjoint components: (i) decentralized client data used for federated training, (ii) a small curated server dataset containing representative samples from known attack classes for semantic alignment and threshold calibration, and (iii) a held-out test set used exclusively for evaluation.

B. Federated Configuration

We simulate a federated system with K clients. To induce realistic heterogeneity, client datasets are generated using a Dirichlet distribution over class labels, resulting in strongly non-IID splits [9]. Each client trains a local embedding model using private data only, while the server coordinates aggregation, maintains semantic prototypes, and performs threshold calibration.

Algorithm 1 COSINE: Federated Semantic Intrusion Detection with Gated Rejection**Require:** Client datasets $\{\mathcal{D}_k\}_{k=1}^K$, semantic descriptions $\mathcal{S}$, embedding model f_θ , number of rounds T , local epochs E **Ensure:** Trained global embedding model f_θ and calibrated class-specific thresholds $\{\tau_c\}$

- 1: Initialize global embedding parameters $\theta^{(0)}$ at the server
- 2: Construct initial semantic class prototypes $\{\mathbf{p}_c^{(0)}\}$ by embedding descriptions from $\mathcal{S}$
- 3: **for** $t = 1$ to T **do**
- 4: Select a subset of participating clients $\mathcal{C}_t \subseteq \{1, \dots, K\}$
- 5: **for all** clients $k \in \mathcal{C}_t$ **in parallel do**
- 6: Receive the current global model parameters $\theta^{(t-1)}$
- 7: Optimize local embedding model on private dataset $\mathcal{D}_k$ for E epochs
- 8: Compute local class-wise embedding centroids $\{\bar{\mathbf{p}}_{c,k}\}$ from learned representations
- 9: Transmit updated model parameters and centroids to the server
- 10: **end for**
- 11: Aggregate client model parameters using weighted Federated Averaging (FedAvg)
- 12: Update global semantic prototypes using exponential moving average aggregation
- 13: Calibrate class-specific similarity thresholds $\{\tau_c^{(t)}\}$ using validation statistics
- 14: **end for**
- 15: **return** final global model f_θ and calibrated thresholds $\{\tau_c\}$

C. Training Protocol

Training proceeds in synchronous federated rounds. In each round, a subset of clients performs local optimization and transmits model updates to the server. Updates are aggregated using weighted federated averaging to account for varying client dataset sizes [22]. Semantic prototypes are updated using exponential moving averages to mitigate drift under heterogeneous client distributions.

D. Evaluation Metrics

We report robust accuracy computed over accepted samples, traffic coverage, and rejection statistics distinguishing between ambiguity-based and low-similarity rejections. These metrics jointly characterize the trade-off between classification reliability and system decisiveness.

V. PERFORMANCE EVALUATION

This section evaluates the effectiveness of COSINE under non-IID federated conditions. Unlike conventional closed-set IDS evaluation, rejected samples are explicitly modeled. Performance is assessed using robust accuracy computed over accepted samples, traffic coverage, and rejection behavior, highlighting the trade-off between reliability and system decisiveness.

A. Overall Robust Performance

TABLE I: Overall Robust Performance on the NSL-KDD Test Set

Training Objective	Robust Accuracy (%)	Coverage (%)	Avg. τ
Margin Only	97.82	92.24	0.374
Semantic Only	95.49	94.55	0.736
Semantic + Margin	95.00	95.19	0.374

Table I summarizes final test-set performance across training objectives. Margin-only training prioritizes precision but exhibits conservative coverage. Semantic-only training increases coverage at the cost of robustness. The combined semantic-margin objective achieves the most balanced operating point. Importantly, these objectives correspond to distinct modeling regimes: margin-only training represents geometry without language, semantic-only training represents language without discrimination, and their combination reflects language-defined decision geometry.

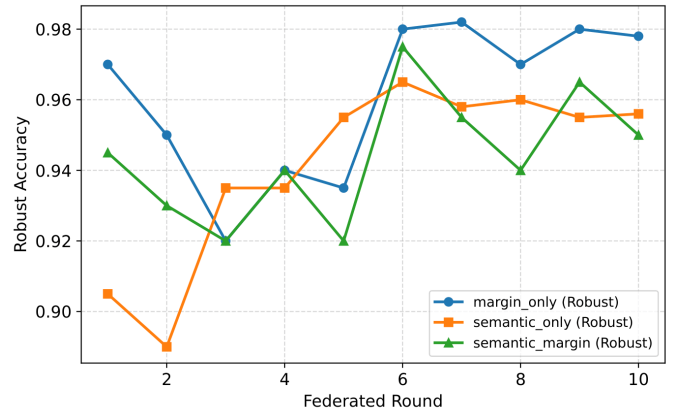
B. Federated Training Dynamics

Fig. 2: Robust test accuracy across federated training rounds.

Figure 2 shows the evolution of robust accuracy over federated rounds. Margin-only optimization converges rapidly but exhibits higher variance in early rounds. Semantic alignment stabilizes learning under client heterogeneity.

Figure 3 illustrates system decisiveness measured as traffic coverage. Semantic objectives gradually relax rejection as rep-

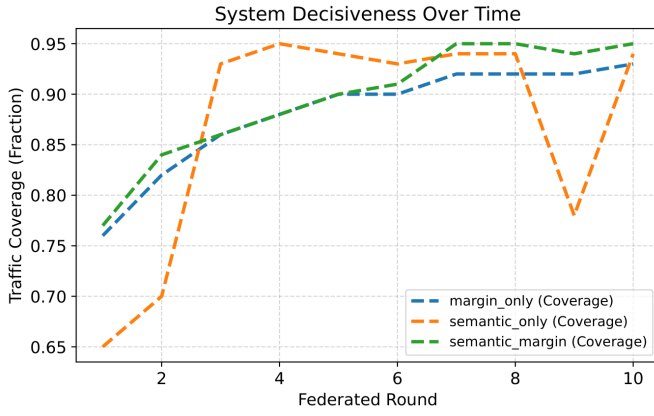


Fig. 3: Traffic coverage across federated training rounds.

representations become more reliable, while margin-only training remains conservative early on.

C. Rejection Behavior Analysis

TABLE II: Robust Split-wise Evaluation (Semantic + Margin)

Split	Robust Accuracy (%)	Coverage (%)
Client Train (Aggregated)	96.46	88.56
Server Train (Known)	95.20	91.30
Full Test Set	95.00	92.10

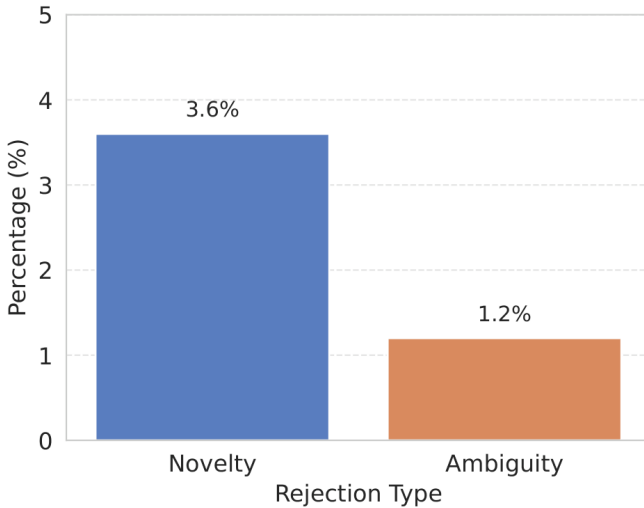


Fig. 4: Breakdown of system decisions on the test set.

Figure 4 decomposes system decisions into confident alerts and rejections. Most rejections arise from low semantic similarity, indicating novelty or out-of-distribution traffic. A smaller fraction is due to ambiguity between competing attack classes, confirming that abstention is driven by uncertainty rather than random confusion.

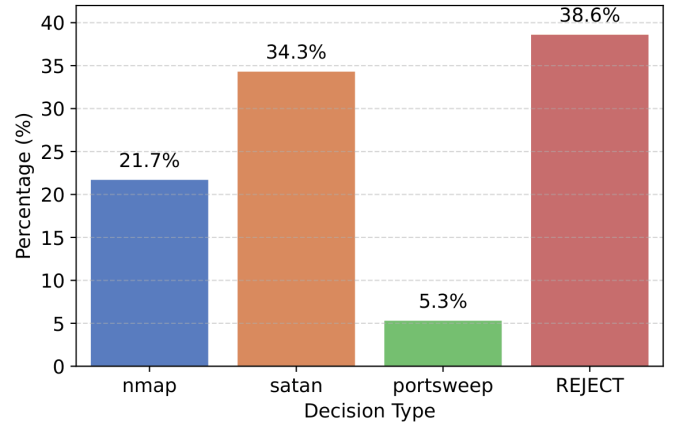


Fig. 5: Zero-shot decision outcomes of the COSINE system (semantic-margin objective).

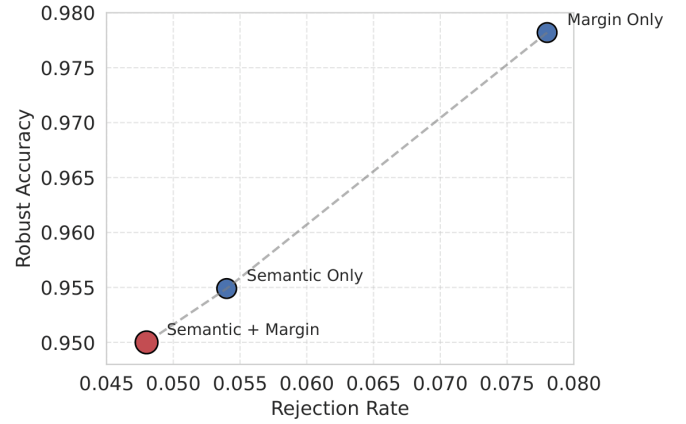


Fig. 6: Trade-off between robust accuracy and rejection rate.

D. Accuracy-Coverage Trade-off

Figure 6 highlights the inherent trade-off between robustness and coverage. Closed-set IDS implicitly operate at full coverage, whereas COSINE explicitly controls abstention through calibrated rejection, preventing overconfident misclassification.

E. Split-wise Robustness

Table II further examines the robustness of COSINE across different evaluation splits under the semantic-margin objective. The close alignment between server-curated training data and the held-out test set indicates that semantic grounding does not overfit to centralized supervision. Slightly reduced coverage on aggregated client data reflects stronger heterogeneity and partial class visibility across clients, reinforcing the need for explicit rejection when operating under non-IID federated conditions.

F. Threshold Calibration

Figure 7 shows that these thresholds are computed relative to semantic prototypes derived from generated descriptions,

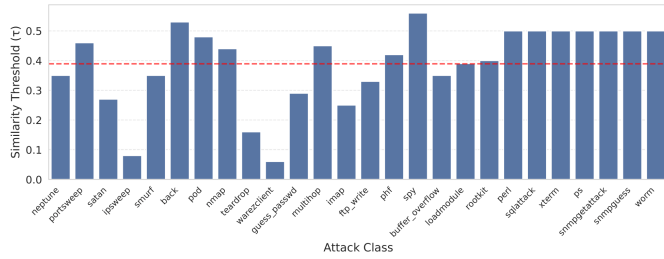


Fig. 7: Distribution of calibrated per-class similarity thresholds (τ).

highlighting that acceptance and rejection are defined in language-induced semantic space rather than probability or feature space.

G. Discussion

Overall, COSINE achieves high robust accuracy while maintaining controlled traffic coverage under severe non-IID conditions. By grounding decisions in semantic similarity and explicitly modeling uncertainty through gated rejection, the system avoids overconfident predictions typical of closed-set federated IDS approaches. These results validate semantic grounding and calibrated rejection as effective design principles for practical federated intrusion detection.

VI. CHALLENGES AND LIMITATIONS

COSINE relies on semantic attack descriptions to construct class prototypes, and inaccurate or ambiguous descriptions may affect similarity calibration. While class-specific thresholds mitigate global overconfidence, rare attack classes with limited samples can still exhibit unstable similarity statistics under strongly non-IID client distributions. In addition, server-side semantic embedding and threshold calibration introduce modest computational overhead, which may impact scalability in large federated deployments.

VII. FUTURE WORK

Future extensions of COSINE include dynamic expansion of the semantic label space, where consistently rejected traffic may be clustered and promoted to new attack classes through human-in-the-loop validation. Incorporating temporal or sequence-aware representations could further improve detection of multi-stage attacks. Adaptive client participation strategies may also enhance robustness under evolving threat landscapes.

VIII. CONCLUSION

This work introduced COSINE, a federated semantic intrusion detection framework that replaces closed-set classification with similarity-based gated rejection. By grounding decisions in semantic embeddings and explicitly modeling uncertainty, COSINE achieves high robust accuracy while maintaining controlled traffic coverage under non-IID conditions. The results demonstrate that semantic grounding and calibrated rejection are effective design principles for practical federated IDS deployment.

REFERENCES

- [1] H.-J. Liao, C.-H. Richard Lin, Y.-C. Lin, and K.-Y. Tung, "Intrusion detection system: A comprehensive review," *Journal of Network and Computer Applications*, vol. 36, no. 1, pp. 16–24, 2013. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1084804512001944>
- [2] R. G. Bace, P. Mell *et al.*, "Intrusion detection systems," 2001.
- [3] Y. He, M. Zhang, X. Yang, J. Luo, and Y. Chen, "A survey of privacy protection and network security in user on-demand anonymous communication," *IEEE Access*, vol. 8, pp. 54 856–54 871, 2020.
- [4] Y.-H. Chiang, L.-H. Chang, and T.-H. Lee, "Federated learning for network traffic classification: Impact of non-iid distribution on model performance," in *Proceedings of the 2023 8th International Conference on Cloud Computing and Internet of Things*, 2023, pp. 71–77.
- [5] U. Yadav and A. Sharma, "Network security in evolving networking technologies: developments and future directions," *Evolving Networking Technologies: Developments and Future Directions*, pp. 75–95, 2023.
- [6] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Processing Magazine*, vol. 37, no. 3, pp. 50–60, 2020.
- [7] J. Li, X. Tong, J. Liu, and L. Cheng, "An efficient federated learning system for network intrusion detection," *IEEE Systems Journal*, vol. 17, no. 2, pp. 2455–2464, 2023.
- [8] E. Fedorchenko, E. Novikova, and A. Shulepov, "Comparative review of the intrusion detection systems based on federated learning: Advantages and open challenges," *Algorithms*, vol. 15, no. 7, p. 247, 2022.
- [9] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-iid data," *arXiv preprint arXiv:1806.00582*, 2018.
- [10] X.-Y. Zhang, G.-S. Xie, X. Li, T. Mei, and C.-L. Liu, "A survey on learning to reject," *Proceedings of the IEEE*, vol. 111, no. 2, pp. 185–215, 2023.
- [11] A. Hliaoutakis, G. Varelakis, E. Voutsakis, E. G. Petrakis, and E. Milios, "Information retrieval by semantic similarity," *International journal on semantic Web and information systems (IJSWIS)*, vol. 2, no. 3, pp. 55–73, 2006.
- [12] G. A. Miller and W. G. Charles, "Contextual correlates of semantic similarity," *Language and cognitive processes*, vol. 6, no. 1, pp. 1–28, 1991.
- [13] T. Paz-Argaman, R. Tsarfaty, G. Chechik, and Y. Atzmon, "Zest: Zero-shot learning from text descriptions using textual similarity and visual summarization," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 569–579.
- [14] R. Dasgupta and P. Mitra, "Large language model-based federated zero-shot learning for intrusion detection in smart grids," in *Artificial Intelligence Applications and Innovations*, I. Maglogiannis, L. Iliadis, A. Andreou, and A. Papaleonidas, Eds. Cham: Springer Nature Switzerland, 2025, pp. 331–344.
- [15] R. S. C. Kone, "Exploring the dynamics and countermeasures of label-flipping attacks in intrusion detection systems," Ph.D. dissertation, Morgan State University, 2024.
- [16] S. Rahman, S. Khan, and F. Porikli, "A unified approach for conventional zero-shot, generalized zero-shot, and few-shot learning," *IEEE Transactions on Image Processing*, vol. 27, no. 11, pp. 5652–5667, 2018.
- [17] L. Zhang, T. Xiang, and S. Gong, "Learning a deep embedding model for zero-shot learning," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 2021–2030.
- [18] X. Xu, C. Liu, Q. Feng, H. Yin, L. Song, and D. Song, "Neural network-based graph embedding for cross-platform binary code similarity detection," in *Proceedings of the 2017 ACM SIGSAC conference on computer and communications security*, 2017, pp. 363–376.
- [19] Q. Li, B. He, and D. Song, "Model-contrastive federated learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2021, pp. 10 713–10 722.
- [20] N. Charoenphakdee, Z. Cui, Y. Zhang, and M. Sugiyama, "Classification with rejection based on cost-sensitive classification," in *International Conference on Machine Learning*. PMLR, 2021, pp. 1507–1517.
- [21] S. Revathi and A. Malathi, "A detailed analysis on nsl-kdd dataset using various machine learning techniques for intrusion detection," *International Journal of Engineering Research & Technology (IJERT)*, vol. 2, no. 12, pp. 1848–1853, 2013.
- [22] H. Wang, M. Yurochkin, Y. Sun, D. Papailiopoulos, and Y. Khazeni, "Federated learning with matched averaging," *arXiv preprint arXiv:2002.06440*, 2020.